COURSE PROJECT PROPOSAL

AI-Powered Virtual Showroom Digital Twin

IoT Predictive Marketing Analytics · Apple Vision Inspired · Spatial Computing

Field	Details
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Project Type	Group Project (Group 12)
Submission Date	May 20, 2025
Course	Cyber Physical Systems
Instructor	Dr. Fatehi
Inspired By	Apple Vision Pro · Amazon Go · Tesla Showroom

1. Problem Statement

Customers today cannot physically test or compare products when shopping online or in large stores. Traditional retail misses' real-time engagement data by the time managers react, the customer is already gone.

This project builds an AI Powered Virtual Showroom Digital Twin a low cost IoT system that tracks customer proximity to products in real time, scores engagement using a machine learning model, and delivers smart product recommendations through a live spatial dashboard. The physical world and digital world are linked live this is the definition of a Digital Twin.

This directly mirrors how Apple Vision Pro creates a spatial computing layer over reality, how Amazon Go tracks customer product interaction, and how Tesla uses data driven showroom analytics.

2. Proposed Solution

A product (e.g. a mini car model or phone box) sits on a table. An ultrasonic sensor detects when a person approaches. The ESP32 microcontroller captures distance data and streams it over Wi-Fi to a Python Streamlit dashboard on a laptop.

The dashboard acts as the virtual showroom updating live with engagement scores, dwell time, and AI-generated recommendations. When a recommendation fires, an LED beside the suggested product lights up. All data is stored in SQLite for trend analysis.

3. Apple Vision Pro Inspiration

Apple Vision Pro uses spatial computing cameras, LiDAR, eye-tracking, and IMUs to overlay digital information onto the real world. This project simulates the same concept at student scale:

Apple Vision Pro Concept	This Project's Equivalent
Spatial computing layer	Virtual showroom dashboard (Streamlit)
LiDAR / depth sensing	Ultrasonic HC-SR04 distance sensor
Eye / hand interaction detection	Proximity + dwell time tracking
Real world + digital overlay	Physical product + live digital twin
Smart AI experience	Scikit-learn engagement + recommendation model
Always-on sensors	ESP32 continuous Wi-Fi data stream

Simulating a concept at student scale is completely standard in academic and industry research. Many published papers on smart stores, autonomous systems, and digital twins use this exact approach.

4. System Components

4.1 Sensors

Sensor	Role	Cost (CAD)
HC-SR04 Ultrasonic Sensor	Detect customer proximity to product (simulates LiDAR)	~$3
Optional: Webcam	Basic image capture for engagement logging	Already owned

4.2 Actuators

Actuator	Role	Cost (CAD)
RGB LED (x4)	Light up beside recommended product	~$2
Buzzer	Alert on high engagement anomaly event	~$2

4.3 Wireless Communication

- Platform: ESP32 microcontroller (Wi-Fi built in)
- Protocol: HTTP POST or MQTT where ESP32 sends sensor readings to laptop server
- Dashboard: Streamlit Python app running locally on laptop via Wi-Fi
- No external cloud needed, fully local, privacy-preserving (Apple-style)

4.4 Data Storage

- SQLite: Stores all sensor events with timestamp, distance, engagement score, recommendation
- CSV export: Pandas saves daily session logs for offline analysis
- Real-time: Streamlit reads live from SQLite every 2 seconds, no page refresh needed

5. Hardware Budget

Component	Qty	Est. Cost (CAD)
ESP32 Dev Board	1	$10
HC-SR04 Ultrasonic Sensor	1	$3
RGB LED	2	$2
Buzzer	1	$2
Breadboard + Jumper Wires	1 set	$5
USB Cable (ESP32 power)	1	Already owned
Laptop (dashboard)	1	Already owned
TOTAL		~$22 CAD

All components available on Amazon.ca with 2-day shipping. No soldering required breadboard connections only.

6. Software Stack

Layer	Technology	Purpose
Sensor firmware	Arduino IDE (C++)	Read ultrasonic, send Wi-Fi POST
Backend server	Python Flask (local)	Receive ESP32 data, write to SQLite
Database	SQLite + Pandas	Store & query all engagement events

Layer	Technology	Purpose
ML model	Scikit-learn	Engagement scoring + product recommendation
Dashboard	Streamlit	Futuristic live spatial UI
Visualization	Plotly / Altair	Live charts, heatmaps, trend graphs

7. Data Science & ML Layer

The intelligence layer is what separates this from a basic sensor project and makes it directly relevant to data science roles at Amazon, Apple, and Google:

DS Feature	Technique	Job Market Relevance
Engagement scoring	Distance × dwell time(0 to 100 score)	Customer analytics, A/B testing
Product recommendation	Rule based + KNN classifier	Amazon recommender systems
Anomaly detection	Z score on dwell time	Google anomaly detection pipelines
Session heatmap	Pandas pivot + Plotly heatmap	Apple Store spatial analytics
Trend prediction	Linear regression on hourly traffic	Retail forecasting models
Privacy-first logging	Local SQLite only, no cloud	Apple's on-device ML philosophy

8. Demo Plan

The demo runs on a small table. Total setup time: under 10 minutes.

Step	What Happens	What Audience Sees
1	Person walks near Product A on table	Ultrasonic sensor triggers in real time
2	ESP32 sends distance + time via Wi-Fi	Dashboard card: Customer Detected
3	ML model scores engagement	Engagement Score: 82% appears live
4	Recommendation model fires	Recommended: Product B highlighted on dashboard
5	LED beside Product B glows green	Physical LED lights up, spatial connection
6	Data saved to SQLite	Session log graph updates with new data point
7	Dwell time anomaly detected	Buzzer sounds + dashboard shows anomaly alert

9. Project Timeline

Week	Milestone
Week 1	Purchase ESP32 + sensors. Set up Arduino IDE. Test ultrasonic distance readings.
Week 2	Build Wi-Fi HTTP pipeline: ESP32, Flask server, SQLite database.
Week 3	Build Streamlit dashboard: live KPIs, engagement score card, recommendation output.
Week 4	Train ML model (KNN + scoring logic). Integrate LED actuator with recommendations.
Week 5	Add Plotly heatmaps, anomaly detection, trend chart. Polish UI for futuristic look.
Week 6	Full demo rehearsal. Final documentation and submission.

10. Industry & Job Market Relevance

Company	Real Product This Mirrors	Skills Demonstrated
Apple	Vision Pro spatial computing, Core ML on-device privacy	Edge AI, sensor fusion, spatial UX
Amazon	Amazon Go customer tracking, Personalize recommendations	Real-time pipelines, ML recommenders
Tesla	Showroom engagement analytics, over-the-air data systems	IoT data streams, dashboard design
Google	Retail AI, anomaly detection in Cloud operations	Time-series ML, Vertex AI concepts
Any Retailer	Smart shelf analytics, foot traffic heatmaps	Full-stack IoT + DS integration

11. Conclusion

The AI Powered Virtual Showroom Digital Twin combines IoT sensors, real-time analytics, machine learning, and wireless communication to create a smart retail experience. By connecting physical products with a live digital dashboard, the project demonstrates how modern companies like Apple, Amazon, and Tesla use AI driven customer engagement systems. The project is affordable, practical to build at home, and showcases industry relevant skills in IoT, Python, machine learning, dashboards, and real time data systems.

This project shows we can connect physical hardware to intelligent software systems, the core skill of modern IoT data science.